

FSx for NetApp ONTAP

Summary of the Lab

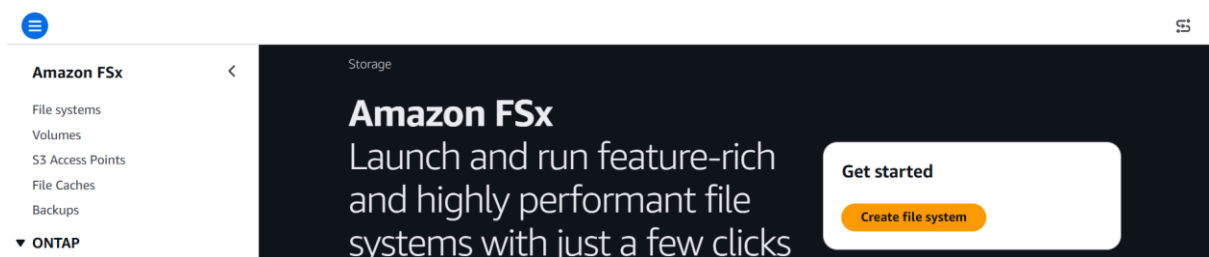
This lab demonstrated how to deploy and configure Amazon FSx for NetApp ONTAP with two Amazon Linux EC2 instances in different Availability Zones. A shared NFS volume was created, mounted on both instances, and tested by creating and accessing files from each server, confirming shared storage functionality. The lab also covered configuring security groups, creating an SVM and data volume, and performing resource cleanup by deleting the volume, SVM, file system, and EC2 instances.

Prerequisites

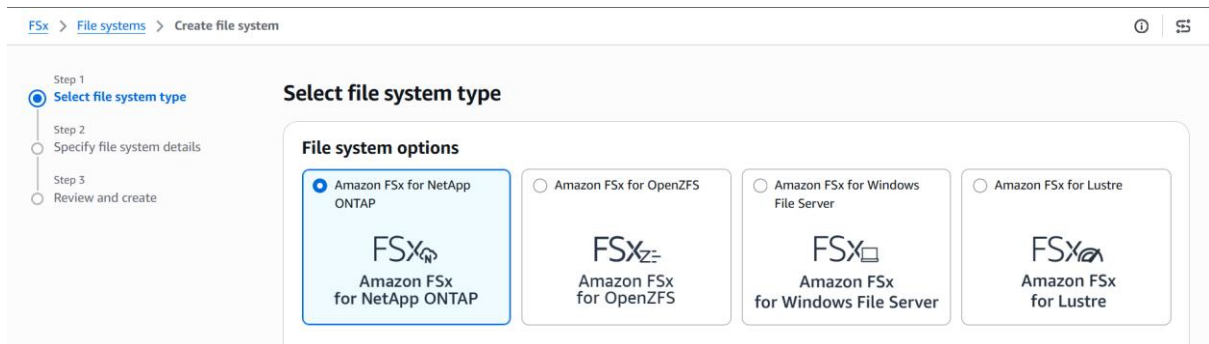
- Create two Security Groups: one for the EC2 servers and one for the FSx ONTAP file system.
- Allow SSH (port 22) to the EC2 Security Group so you can connect to the instances.
- Allow NFS ports **111** and **2049** in the FSx ONTAP Security Group, with access only from the EC2 Security Group.
- Launch two Amazon Linux EC2 instances in different Availability Zones and attach the EC2 Security Group.
- Keep an SSH key pair (.pem file) ready to connect to both EC2 instances.
- Ensure you have permissions to create Amazon FSx for NetApp ONTAP resources in your AWS account.

To Begin with the Lab

- Open the AWS Management Console and navigate to the **Amazon FSx** service.



- Select **Amazon FSx for NetApp ONTAP** and start creating a new file system.



- Choose **Standard Create** to configure the file system manually.

FSx > File systems > Create file system

Step 1 Select file system type
Step 2 **Specify file system details**
Step 3 Review and create

Specify file system details

Creation method

☐ Quick create
Use recommended best-practice configurations. Most configuration options can be changed after the file system is created.

☒ **Standard create**
You set all of the configuration options, including specifying performance, networking, security, backups, and maintenance.

- Enter a name for the file system and select **Single-AZ** deployment.
- Set the storage capacity to **1 TB** and keep the default performance settings.

File system details

File system name - optional | [Info](#)

new-ONTAP

Maximum of 256 Unicode letters, whitespace, and numbers, plus + - = . _ : /

Deployment type | [Info](#)

☐ Multi-AZ 2 (Recommended)
Recommended for workloads that need up to 6 GBps or 200K IOPS. Multi-AZ file systems have a standby server in a second Availability Zone (AZ), protecting your data against instance failure and AZ disruption.

☐ Single-AZ 2
Recommended for workloads that need over 6 GBps of throughput or over 200K IOPS. To protect against instance failure and AZ disruption, you can replicate workloads on Single-AZ filesystems to a filesystem in another AZ.

☐ Multi-AZ 1

☒ **Single-AZ 1**

SSD storage capacity | [Info](#)

1024 GiB

Minimum 1,024 GiB; Maximum 196,608 GiB (192 TiB).

- Select your VPC, choose the previously created FSx Security Group, and select the Availability Zone.

Network & security

Virtual Private Cloud (VPC) | [Info](#)
Specify the VPC from which your file system is accessible.

vpc-099792461c7d0a616 (CIDR: 172.31.0.0/16)

VPC Security Groups | [Info](#)
Specify VPC Security Groups to associate with your file system's network interface.

Choose VPC security group(s)

sg-0bb9f6de77df9ad40 (FSx-ONTAP-SG) X

Subnet | [Info](#)
Specify the subnet in which your file system's network interface resides.

subnet-0b5783ef03796c671 (us-east-1a | use1-az4)

Network type
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with the VPC you specify.

☒ **IPv4**
Your file system can communicate only over the IPv4 addressing protocol.

☐ Dual-stack
Your file system can communicate over IPv4, IPv6, or both addressing protocols.

- Keep the default encryption settings and create a new Storage Virtual Machine (SVM).

Default storage virtual machine configuration

Storage virtual machine name | [Info](#)
The name of the SVM's root volume will be this name with '_root' appended to the end.

ONTAP-New

Maximum of 47 alphanumeric characters, plus . - _ . Must begin with a letter or underscore

SVM administrative password | [Info](#)
Password for this SVM's "vsadmin" user, which you can use to access the ONTAP CLI or REST API. You can provide a password later if you don't provide one now.

☒ Don't specify a password
☐ Specify a password

Volume security style | [Info](#)
The security style of the volume determines whether preference is given to NTFS or UNIX ACLs for multi-protocol access.

Unix (Linux) ▼

Active Directory | [Info](#)
Joining an Active Directory enables access from Windows and MacOS clients over the SMB protocol.

☒ Do not join an Active Directory
☐ Join an Active Directory

- Create a data volume (for example, 500 MB) with **Read/Write** access.

Default volume configuration

Volume name

Vol_1

Maximum of 203 alphanumeric characters, plus _ .

Volume style

☒ FlexVol (recommended)
FlexVols are the standard ONTAP volume type that can be as large as 300 terabytes.

☐ FlexGroup
FlexGroups are composed of multiple hidden volumes called constituents and can be as large as 20 petabytes.

Volume size
Minimum 20 MiB; Maximum 314,572,800 MiB

500 MiB ▼

Volume type
Select whether you want your file system's first volume to be a Read-Write (RW) volume or a read-only Data Protection (DP) volume, which is used with SnapMirror.

☒ Read-Write (RW)
☐ Data Protection (DP)

- Disable snapshots and backups for this lab, then create the file system.

Junction path
The location within your file system where your volume will be mounted.

/vol1

Storage efficiency
Select whether you would like to enable ONTAP storage efficiencies on your volume: deduplication, compression, and compaction.

☐ Enabled (recommended)
☒ Disabled

Snapshot policy | [Info](#)
The snapshot policy of the volume determines the schedule on which snapshots are automatically taken of your volume.

None ▼

Default Volume SnapLock Configuration

SnapLock Configuration
Store files using a write-once-read-many (WORM) model to prevent data from being deleted or overwritten for a user-defined period.

☐ Enabled
☒ Disabled

▼ **Backup and maintenance *optional***

Daily automatic backup | [Info](#)
Amazon FSx can protect your data through daily backups

☐ Enabled
☒ Disabled

Weekly maintenance window | [Info](#)
When patching needs to be performed, Amazon FSx performs maintenance on your file system only during this window.

☒ No preference
☐ Select start time for 30-minute weekly maintenance window

- Wait until the FSx ONTAP file system becomes **Available**.

File systems (1)

Filter file systems by property or tag

< 1 >

⚙

Attach

Actions ▼

Create file system

	File system name ▼	File system ID ▼	File system type ▼	Status ▼	Deployment type ▼	Storage class ▼	Storage capacity ▼	Through capacity
☐	new-ONTAP	fs-0d120cbe9c0cb7909	ONTAP	Creating	Single-AZ 1	SSD	1,024 GiB	128 MB/

- Open the created volume and view the **Linux mount instructions** provided by AWS.

Vol_1 (fsvol-0cb3114b2041b4fd8)

Attach

Actions ▼

Create volume

Summary Volume ID fsvol-0cb3114b2041b4fd8 Volume name Vol_1 UUID 6e91c4c1-81a3-11f1-8d12-6d2166b08be2 Creation time 2026-07-17T11:00:57+05:30 Lifecycle state Created Volume type ONTAP ONTAP volume type RW Junction path /vol1 Tiering policy name SNAPSHOT_ONLY SnapLock type Disabled			
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Vol_1

Summary

Volume ID

fsvol-0cb3114b2041b4fd8

Volume name

Vol_1

UUID

6e91c4c1-81a3-11f1-8d12-6d2166b08be2

From Linux instances

Prerequisites

Attach instructions

1. Create or select a Linux EC2 instance in the same AWS VPC as your file system.
 2. Open an SSH client and connect to your EC2 instance. [Find out how to connect.](#)

1. Open a terminal
 2. Create a new directory on your EC2 instance (for example, /fsx)
 `sudo mkdir /fsx`
 3. Mount your volume
 `sudo mount -t nfs svm-02595837fe4e45e75.fs-0d120cbe9c0cb7909.fsx.us-east-1.amazonaws.com:/vol1 /fsx`

Attach

Actions ▼

Create volume

SnapLock type

Disabled

- Connect to both EC2 instances using SSH with your key pair
- Give the give the permission for the first and then connect to the server 1 and 2

- Switch to the root user on each EC2 instance using `sudo -i`.
- Create the mount directory by running the command provided in the FSx console.
- Mount the FSx ONTAP NFS volume on both EC2 instances using the AWS-generated mount command.
- Verify that the mount was successful by listing the contents of the mounted directory.

- Created a file in the mounted FSx ONTAP directory on the first EC2 instance and verified it was accessible from the second instance. Then created another file/folder on the second instance and confirmed it appeared on the first, validating that both EC2 instances share the same Amazon FSx for NetApp ONTAP storage.

- After testing, unmount the volume if required and terminate both EC2 instances.
- Delete the FSx data volume, then delete the Storage Virtual Machine (SVM).
- Finally, delete the FSx ONTAP file system to avoid unnecessary AWS charges.